Audit of the SV efficiency proof: proof-check and literature scan (October 2026)

Source audited: `sv_efficiency_proof.md` (the source of `sv_efficiency_proof.tex`, which is not in the repository); rendered PDF: `sv_efficiency_proof_md.pdf`. Not refereed.

Part A. 1. Claim (Theorem 2) and labelled assumptions

For $\ell\in\{\mathrm{Res}_A,-\tilde S\}$, $\eta_{\ell,K}^2\le C\sum_{K'\subset\tilde\omega_K}[|\ell|_{,\tilde\omega_{K'}}^2+\mathrm{osc}_{K'}^2]$ and $\eta_\ell\le C(|\ell|_+\mathrm{osc})$, with $C=C(d,\vartheta)$ independent of $\varepsilon,\sigma,b,c,h,A,u_h$.

Assumptions used: H1 (mesh), H2 (constant ε), H3 (σ, c_b), H4 (inflow in Γ_D), H6 ($S=(A\nabla\cdot,\nabla\cdot)$, A cellwise constant), A1 (RTN$_1$ inverse), A2 (SV20 L4.1, plus the Neumann shift [EV15]), A3 (squeezed bubbles), A4 (I_c), A5 (L^2-bounded commuting projection, Licht 2019), A6 (finite overlap).

Part A. 2. Step table

	explicit	OK				
						v
_1(K)$	$C(d,\vartheta)$	OK. Minor: "$m_K=\sigma^{-1/2}$" holds only for $\theta\le1/\pi$; $m_K\le\sigma^{-1/2}=h\varepsilon^{-1/2}\theta$ is what is used, and it holds always				

	$	**ψ_a in the local test space**	$C(1+\theta^2)\le2C$	**G1**: false with the §1 definition ($H^1_0(\omega)$) at compatible Neumann vertices		

Status of the fixes (18 Oct 2026)

- **G1:** fixed. The local test space is $H^1_*(\omega)$ throughout `sv_efficiency_proof.md` (§1, A6, Lemma H(i), Lemma D). No code change: `localized_error` already used H^1_*.
- **G2:** fixed.
  - Corollary 3, Theorem 4.3 and Proposition 5.3 now state $(1 + c_b)^2$.

  - New Corollary 3': the weighted measure $E_c = \max\{1, c_b\}|||e||| + \|B_A'e\|_* + \|S^\sim\|_*$ has a guaranteed bound $\eta_{E,c} = (M-1)\eta + \eta_E$, with a two-sided constant linear in c_b.
  - Code: `eta_E_c`, `eta_aug_c` and `c_b_weight` in `sv_estimator`; rows `err_E_c` and `eff_E_c_sv`; test `test_weighted_augmented`.
  - Smoke check: at $c_b = 1000$, $\eta_{E,c}/E_c \approx 1.2c_b$, against $1.9–3.4c_b$ for η_E/E.
- **G3:** fixed.
  - The patch data are the commuting RTN_1 interpolant $\Pi_{RT}(\psi_a D\nabla u_h)$ (`rt1_data`). It is identical to the old data when $D\nabla u_h$ is constant per cell (checked to $4\cdot10^{-16}$) and changes them by 2.6 % for smooth ε at contrast 10^4.
  - The proof of Theorem 6.2 is rewritten with $\hat r_\ell = r_\ell + \mathrm{div}_h(D\nabla u_h)$.
  - Li 2026 is cited for the impossibility without quasi-monotonicity.
- **G4:** fixed. Proposition 5 now states the two-sided bound at every iterate, for every γ, without $C_2\gamma < 1$, because the extra term was counted twice. What the stopping rule adds is $E(u^i) \le (1+\gamma)\eta_{disc}$. The comparison with the discrete solution is stated as open. Smoke check: η_E/E in 2.9–6.0 at iterates 1–2000, even with η_{alg}/η_{disc} up to 0.65. Test `test_iterate_efficiency`.

Part A. 3. Verdict

- **Theorem 2:** correct, with one fillable gap (G1). It also rests on two cited ingredients: A5, and the Neumann version of A2.
- **Corollary 3, Theorem 4.3, Proposition 5.3:** the constant is wrong (G2). The statements hold with $(1+c_b)^2$.
- **Proposition 5.3:** correct, but under a superfluous hypothesis (G4).
- **Theorem 6.2:** gap (G3); the efficiency for variable ε is not established as stated.
- **Theorem 1, Lemma L, Theorem 4.1–4.2, Lemma V, Theorem 6.1:** correct.

Part A. 4. Gaps and fixes

G1 (local test space at Neumann boundaries).

- **Where.** §1 defines $|\ell|_{*,\omega}$ with $H^1_0(\omega)$. Two steps need test functions that do not vanish on Γ_N:
  - Lemma H(i) tests with ψ_a at compatible Neumann vertices;
  - Lemma D's Neumann face term tests with $b_F^\theta\mu$ on $F\subset\Gamma_N$.
- **Fix.** Use the §9 space $H^1_*(\omega)=\{v\in H^1(\omega):v=0\text{ on }\partial\omega\setminus\Gamma_N\}$ throughout. A6 and the monotonicity in domain still hold, by zero extension into H^1_D.

G2 (c_b dependence).

- **Why.** Remark 1.2 loses $\max\{1,c_b\}$ in each direction: $|||e|||_\oplus\le\kappa|\mathrm{Res}|_*$ and $|\mathrm{Res}|_*\le\max\{1,c_b\}|||e|||_\oplus$. Hence $\eta_E\le C\kappa\max\{1,c_b\}(E+\mathrm{osc})$.
- **Which statements.** Corollary 3, Theorem 4.3 and Proposition 5.3 are true with $C(1+c_b)^2$, not $C(1+c_b)$. This is irrelevant when $c_b=1$ (every constant-coefficient example). For AJ27 Ex. 4.3 ($c_b\approx8\cdot10^4$) it is $\approx7\cdot10^{9}$ instead of $\approx8\cdot10^{4}$.
- **Sharper option.** Weight the augmented norm, $|||e|||_{\oplus,c}:=\max\{1,c_b\}|||e|||+|B_A'(e,\cdot)|_*$. Then $|\mathrm{Res}|_*\le|||e|||_{\oplus,c}\le(1+2\max\{1,c_b\})|\mathrm{Res}|_*$, and the effectivity constant is linear in c_b.
- **c_b-free option.** Use the energy norm with the full reaction $c-\tfrac12\mathrm{div}\,b$ (§7).

G3 (Theorem 6.2, $D=\Pi_1\varepsilon I+A$). Three uses of "$D\nabla u_h$ constant on K" fail when $D\nabla u_h\in P_1^d$:

1. **Lemma D:** $\ell(b_{Kw})=(r_\ell+\mathrm{div}(D\nabla u_h),b_{Kw})_K$, with $\mathrm{div}(D\nabla u_h)=\nabla\Pi_1\varepsilon\cdot\nabla u_h\ne0$. This one is harmless: replace r_ℓ by $\hat

r_\ell=r_\ell+\mathrm{div}_h(D\nabla u_h)$ in RES and in Lemmas D and G. The bubble argument is unchanged.

2. **Lemma B / A2:** $\psi_aD\nabla u_h\in P_2^d\not\subset$ RTN$_1$. Lemma B gets the remainder $|(I-\Pi_{RT_1})(\psi_aD\nabla u_h)|_K\lesssim h_K|\nabla\Pi_1\varepsilon|_{\infty,K}|\nabla u_h|_K$. This is first order and not controlled by the error. A2 (SV20 L4.1) needs broken RTN$_1$ data.
3. **Why the estimator is probably still efficient, and what the proof loses.** The remainder cancels in the estimator, because $\sum_a\Pi_{RT_1}(\psi_aD\nabla u_h)=D\nabla u_h\in$ RTN$_1$. But the patchwise proof (triangle inequality in §5.1) does not see this cancellation.
4. **Fix.** Use $\Pi_{RT_1}(\psi_aD\nabla u_h)$ as the patch datum, with the commuting RTN$_1$ interpolant, so $\mathrm{div}\,\Pi_{RT_1}=\Pi_1\mathrm{div}$. This keeps the partition of unity exactly and makes Lemmas A, B and G hold verbatim with $\hat r_\ell$.
5. **Code impact.** Reliability is unaffected (the identity in Theorem 1.1 holds for any σ_h). The fix is a change of data in `_solve_patches`. Until then, Theorem 6.2 should say "not proved". The smoke effectivities 3.2–4.3 at contrast 10^4 suggest the estimator is efficient.

G4 (Proposition 5.3).

- **Why the hypothesis is superfluous.** Theorem 2 at an iterate is stated in $|\mathrm{Res}_A(u^i)|_*$, and $\mathrm{Res}(u^i)(v)=B(u-u^i,v)$. So $\eta_E(u^i)\le C(1+c_b)^2(E(u^i)+\mathrm{osc})$ holds at every iterate, for every γ, and without $C_2\gamma<1$. The extra $|\psi_{\rm def}|_{*,\omega_a}$ term in step 1 double-counts what Lemma H(i) already bounds.
- **What stopping actually adds.** Compare the stopped iterate with the exact discrete solution: $E(u^i)\lesssim E(u_h)$. That needs $|||u_h-u^i|||\lesssim\eta_{\rm alg}$, a discrete stability of the AFC system that is not available, since AFC is not monotone. State it as open.

Part A. 5. Optimality

- **Hypotheses actually used:**
  - H1, A1, A3 and A6 everywhere;
  - H2 in Lemmas B, D and G (constant $D\nabla u_h$);
  - H3–H4 only in Remark 1.2 and Corollary 3 (Theorem 2 itself needs neither: it is a statement about functionals);
  - H6 only through D being cellwise constant;
  - A2 in Lemma G case 2; A4 in H(ii) and Lemma L; A5 only in H(ii), second claim.
- **Weaken H3 to its natural form.** H3's "$\sigma=0\Rightarrow c=\tfrac12\mathrm{div}\,b$" can go: Theorem 2 does not use H3. For Corollary 3, define $c_b:=\min\{|c-\tfrac12\mathrm{div}\,b|_\infty/\sigma,\ |c-\tfrac12\mathrm{div}\,b|_\infty C_F^2/\varepsilon\}$, which is what the code does (`friedrichs_constant`).
- **Weaken H2** to ε piecewise constant on T_h. All lemmas hold verbatim with cellwise ε_K; only the patch-contrast constant appears (the G3 issue disappears because $\Pi_1\varepsilon=\varepsilon_K$).
- **Remove A5.** Bound $\eta_{\ell,\rm def}$ directly by the patchwise dual norms via Lemma L: $\psi_{\rm def}$ is a sum of patch functionals. Equivalently, use a broken (patch-local) RT$_0$ reconstruction of $\psi_{\rm def}$, whose efficiency follows from A2 alone. This trades A5 for one more patch solve per vertex. Not checked in detail.
- **Strongest conclusion.** The arguments give local efficiency in H^1_*-dual norms. The local measure in Theorem 4.2 is the sharpest this route allows. The $(1+c_b)^2$ (G2) is the price of $|||\cdot|||_\oplus$ with $\sigma\ll|c-\tfrac12\mathrm{div}\,b|_\infty$.
- **Sharper known tools:**
  - the p-robust patch stability [EV15/EV20] would give p-robustness, irrelevant at P1;
  - Verfürth's squeezed bubbles (A3) are the sharpest known robust lower-bound tool for this norm;
  - for G3, the commuting interpolant fix is the standard route (as in variable-coefficient equilibration, e.g. Vohralík's inhomogeneous-diffusion setting) [from memory, verify].

Part B. Literature scan (update of `litscan_2026-10.md`)

B1. Answer

The core result has no published competitor found: Theorem 2 with Corollary 3, i.e. a guaranteed, ε- and σ-robust equilibrated estimator for every conforming P1 algebraic stabilisation, two-sided in E. The newest AFC a posteriori work is residual-based and not guaranteed: Ahmed–Jha, arXiv 2602.14504, 2026; Jha 2021; Jha, arXiv 2404.02804.

One new result changes the plan for variable ε. Li, arXiv 2609.27817 (August 2026), proves that **no locally computable estimator for conforming approximations of interface problems can be robust in the coefficient contrast M**. At a checkerboard cross-point, the product of the reliability and efficiency constants grows at least like M^{1/4}. So the "natural next step" proposed after gap 7 (jump-robustness without quasi-monotonicity) is impossible for any local estimator, ours included. The right statement is Theorem 6 with the patch contrast, plus robustness under quasi-monotonicity. I read only the abstract; check the full text.

The track is otherwise unchanged. The proof-check corrections G1–G4 are needed before submission.

B2. Frontier (additions to the October scan)

result	scheme	estimate	hypotheses / limits	mechanism	uniform in
Li, arXiv 2609.27817 (2026)	conforming FEM, interface problems	**impossibility:** $C_{rel} \cdot C_{eff} \gtrsim M^{1/4}$ for any local estimator	checkerboard cross-point (non-quasi-monotone)	two instances with equal local data and diverging errors	— (negative result)
Cai, He, Zhang (arXiv 1603.01024); Cai, Cai, Zhang (arXiv 2001.09102)	nonconforming / DG; conforming higher order	robust in jumps	quasi-monotone for conforming efficiency; nonconforming reliability without	modified residuals / generalised Prager–Synge	contrast (under QM)
Licht, Math. Comp. 88 (2019) 607–635	de Rham FE complexes	smoothed projections that commute and **preserve homogeneous BCs on a boundary part**	shape regularity, bounded degree; weakly Lipschitz	smoothing + extension	h
Ahmed–Jha, arXiv 2602.14504 (2026)	AFC (BJK, MC), MUAS, SMUAS, BBK, adaptive 2D	residual estimator; reliable	not guaranteed	residual	—
Ern–Vohralík SISC 35 (2013); Dolejší–Šebestová–Vohralík	nonlinear diffusion; DG	stopping criteria, algebraic flux reconstruction	monotone / Lipschitz	equilibrated fluxes	—
galerkit Prop. 5	AFC, any solver	$E(u^i) \le \eta_{disc} + \eta_{alg}$, two-sided at every iterate (after G4)	—	one global RT_0 solve for η_{alg}	ε, σ, h

B3. Status of the four open points

1. **c_b dependence (G2).** I found no robust estimator in the augmented norm whose constant is independent of $c - \tfrac{1}{2}\operatorname{div} b$ when $\sigma \ll \|c - \tfrac{1}{2}\operatorname{div} b\|_\infty$.
   - The functional (Repin-type) bounds use the full weight $c - \operatorname{div} b \ge 0$ in the norm (arXiv 1601.05310).
   - Verfürth 2005 and Cai & Cai 2025 assume $\|c\|_\infty \le c_b\, \beta$, i.e. the same c_b.
   - **Best fix:** the energy norm with the full reaction weight, or the weighted augmented norm $|||\cdot|||_{\oplus,c}$ of G2, which gives linear dependence.
   - Effort is low and the novelty small, but it is needed for AJ27 Ex. 4.3.
2. **A5 with mixed BCs.** It is now verified at the level of the abstract: Licht 2019 constructs commuting smoothed projections that preserve homogeneous boundary conditions on a fixed boundary part, with uniform bounds. Check in the full text that the bound is in L^2 for the H(div) level. If so, A5 holds and the "[verify]" flags can be removed.
3. **Discontinuous ε without quasi-monotonicity.** This is impossible for local estimators (Li 2026). It is not a gap to target. State Theorem 6 with the patch contrast, and add a remark citing Li.
4. **AFC stopping criteria.** I still found nothing AFC-specific in 2025–2026. After G4, Proposition 5 gives two-sided bounds at every iterate.
   - **The real open question** is to compare the stopped iterate with the exact discrete solution. That needs a discrete stability (an inverse of the nonlinear AFC operator), which fails without monotonicity.

- **Optimal cost** in the sense of [HPSV21] is open for AFC.

B4. Gaps ranked by novelty-to-effort

1. **G2 + G3 + G1 fixes**, so the paper is correct. Effort is low. Novelty: none, but required.
2. **Remove A5** via a patchwise defect (Part A §5). Medium effort; it would make the proof self-contained.
3. **Efficiency of the stopped iterate against the discrete solution** for monotone limiters, if any AFC variant is monotone (BBK-type symmetric?). Novelty is medium, risk high.
4. **Variable ε with the commuting-interpolant data (G3).** The code change is small; the result is an adaptation.

B5. References

- Y. Li, Locally computable error estimators for conforming approximations of interface problems cannot be robust, arXiv 2609.27817 (2026). [abstract]
- M. W. Licht, Smoothed projections and mixed boundary conditions, Math. Comp. 88 (2019) 607–635, arXiv 1710.06868. [abstract]
- M. W. Licht, Smoothed projections over weakly Lipschitz domains, Math. Comp. 88 (2019) 179–210. [abstract]
- N. Ahmed, A. Jha, Adaptive finite elements with algebraic stabilization for convection-dominated transport, arXiv 2602.14504 (2026). [abstract]
- A. Jha, Residual-based a posteriori error estimators for algebraic stabilizations, arXiv 2404.02804; A. Jha, CAMWA 2021, arXiv 2005.02938. [abstracts]
- Cai, He, Zhang, arXiv 1603.01024; Cai, Cai, Zhang, arXiv 2001.09102. [abstracts]
- Functional a posteriori control for reaction–convection–diffusion, arXiv 1601.05310. [abstract]
- D. Cai, Z. Cai, arXiv 2107.06341. [arXiv version read earlier]
- A. Ern, M. Vohralík, SISC 35 (2013), HAL. [abstract]
- I. Smears, M. Vohralík, ESAIM: M2AN 54 (2020); arXiv 1812.06678. [read]
- Allendes, Barrenechea, Rankin, SISC 39 (2017); Ern, Stephansen, Vohralík, JCAM 234 (2010). [read]